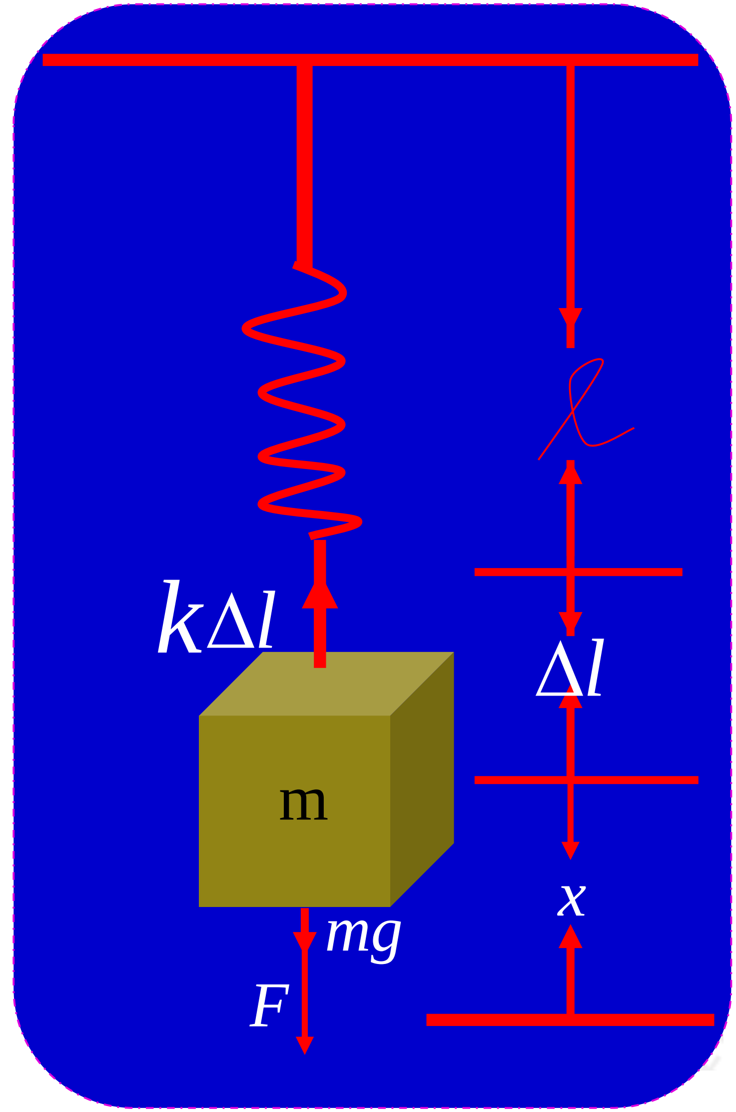
§3.5 二阶微分方程在机械振动中应用

3.5.1 机械振动

设有一弹簧，上端固定，下端挂一质量m的物体，求其振动规律。

物体受4个力的作用：（1）重力；（2）弹性力；（3）空气阻力；（4）外力。



根据运动定律得到方程

$$m\frac{d^2x}{dt^2}=mg-k(\Delta l+x)-c\frac{dx}{dt}+F(t)$$

$$m\frac{d^2x}{dt^2}+c\frac{dx}{dt}+kx=F(t)$$

1、 无阻尼自由运动——无空气阻力和外力作用

$$m\frac{d^2x}{dt^2}+kx=0 \quad 或 \quad \frac{d^2x}{dt^2}+\omega_0^2x=0, \quad \omega_0^2=\frac{k}{m}$$

$$x(t)=c_1\cos\omega_0 t+c_2\sin\omega_0 t$$

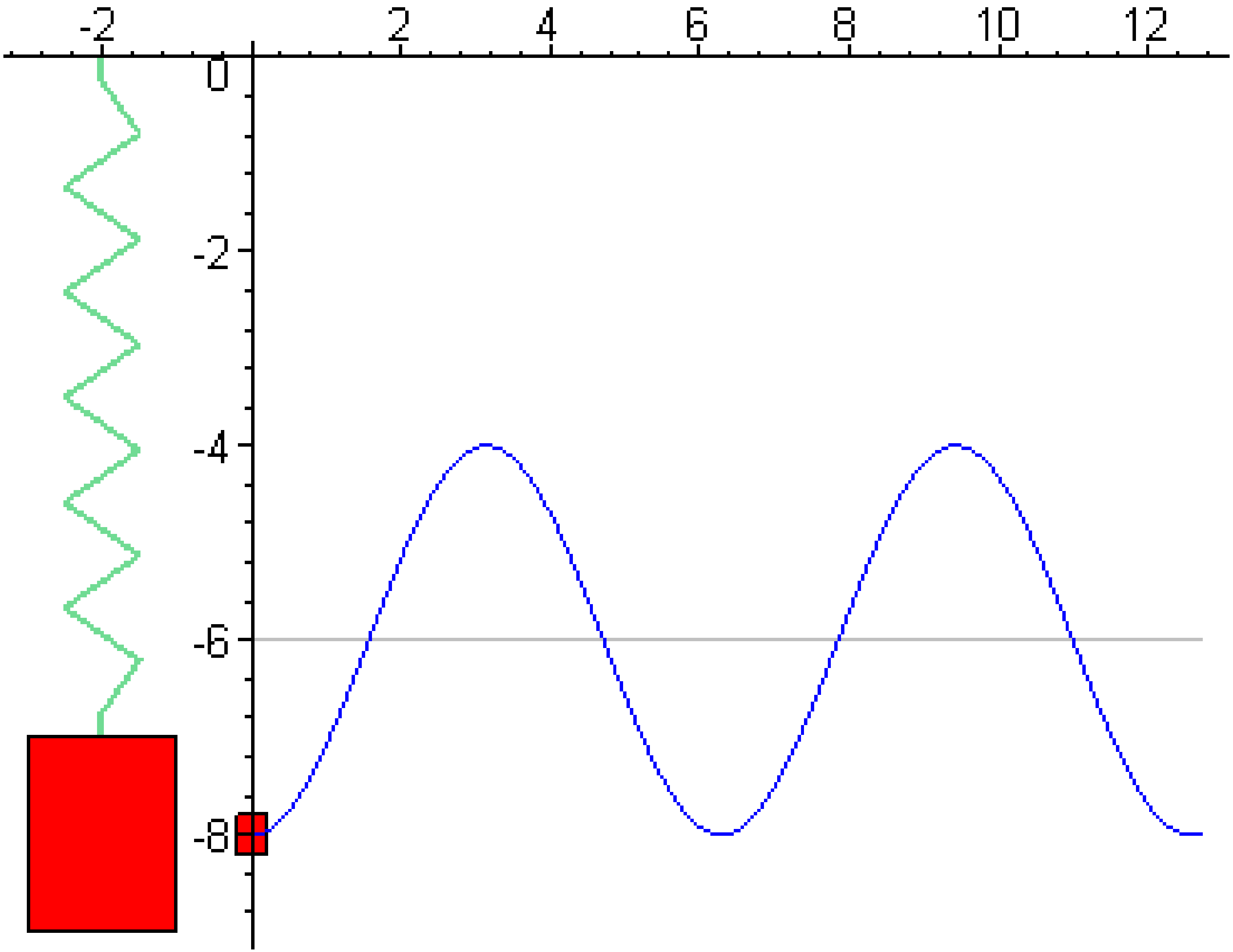
$$x(t)=\sqrt{c_1^2+c_2^2}\left(\frac{c_1}{\sqrt{c_1^2+c_2^2}}\cos\omega_0 t+\frac{c_2}{\sqrt{c_1^2+c_2^2}}\sin\omega_0 t\right)$$

$$=A(\sin\theta\cos\omega_0 t+\cos\theta\sin\omega_0 t)$$

$$=A\sin(\omega_0 t+\theta)$$ （称为简谐振动）

振幅$A=\sqrt{c_1^2+c_2^2}$，周期$T=\dfrac{2\pi}{\omega_0}$，初相位$\theta=\arctan\dfrac{c_1}{c_2}$

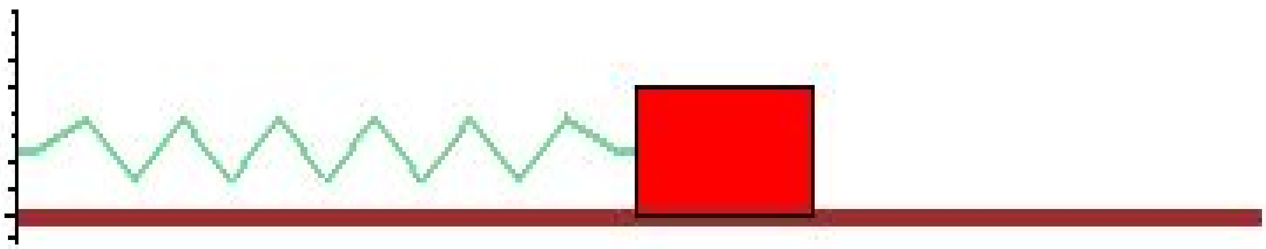
无阻尼自由运动的演示



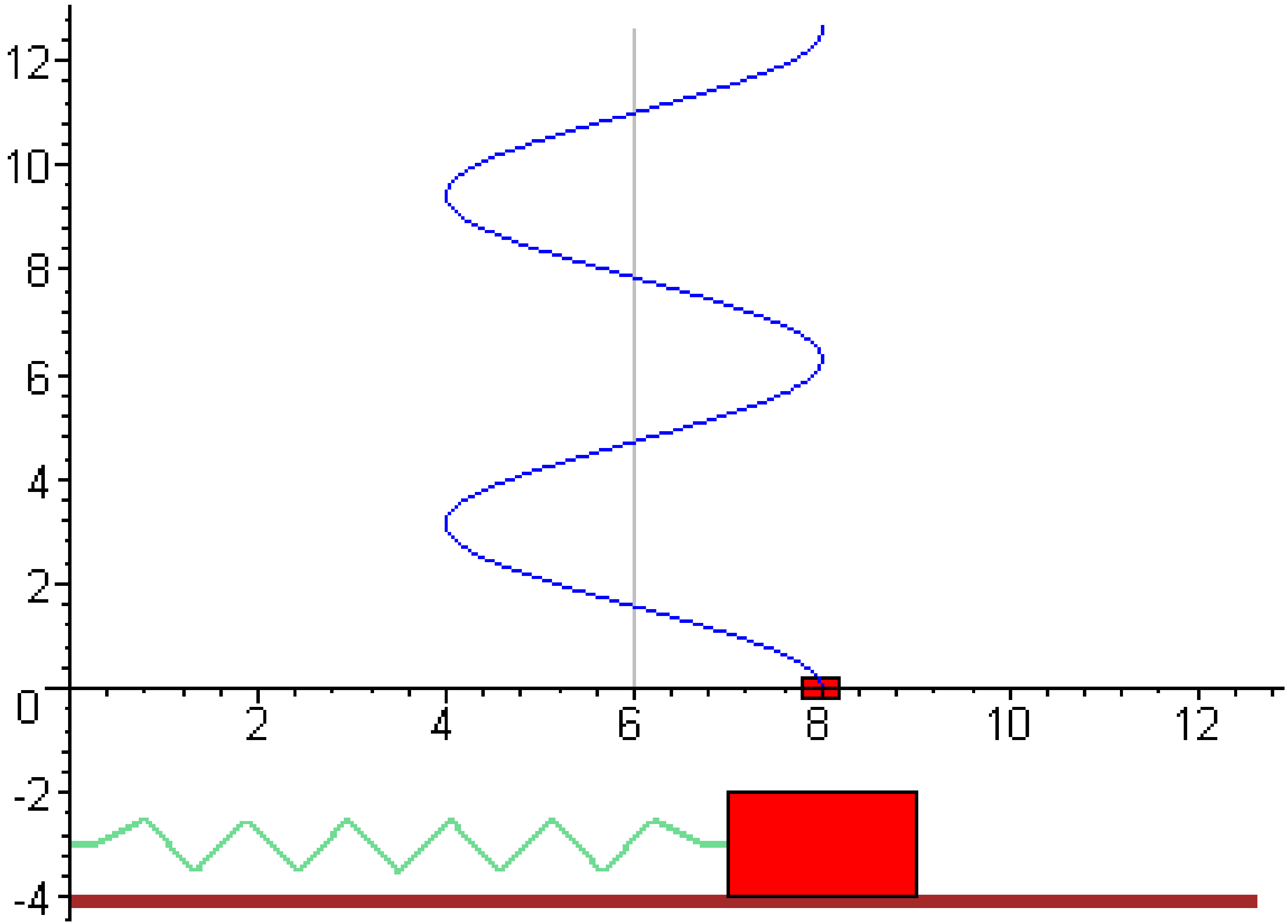
水平无阻尼自由运动

问题

一个弹性系数为k的弹簧左端固定，右端连接一个质量为m的滑块，放在一个光滑的水平面上，给出一定的初始位移后让其运动，求该弹簧质量系统的运动规律。



水平无阻尼自由运动的动画演示



2、 有阻尼的自由振动——有空气阻力而无外力作用

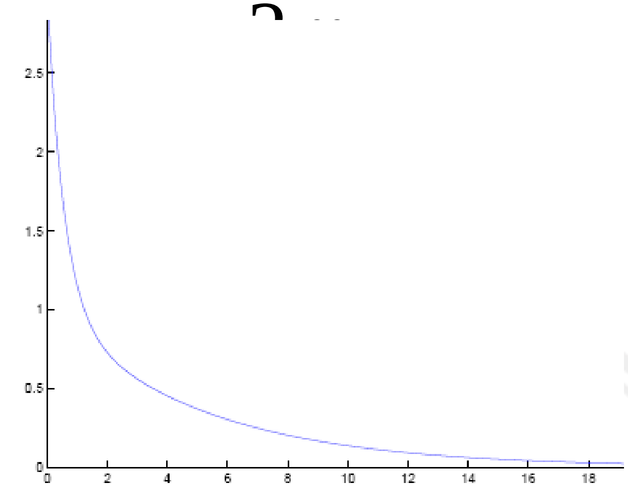
$$m\frac{d^2x}{dt^2}+c\frac{dx}{dt}+kx=0 \quad m\lambda^2+c\lambda+k=0$$

特征根为：$\lambda_1=\dfrac{-c+\sqrt{c^2-4km}}{2m},\lambda_2=\dfrac{-c-\sqrt{c^2-4km}}{2m}$

三种情况：

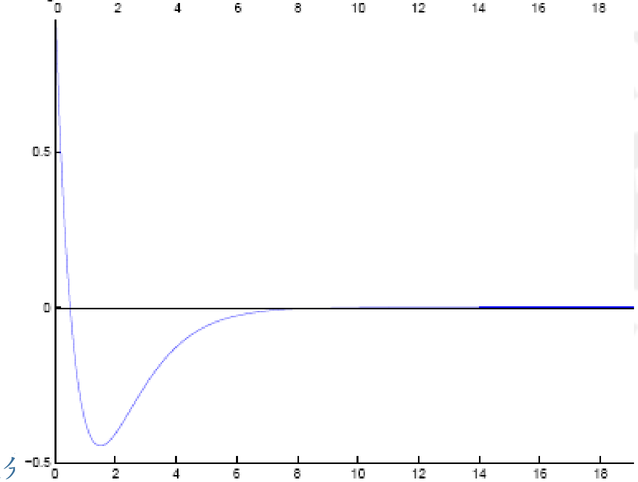
(1) $c^2-4km>0$，两负根

大阻尼：$x(t)=c_1e^{\lambda_1 t}+c_2e^{\lambda_2 t}$



（2）$c^2-4km=0$，重根

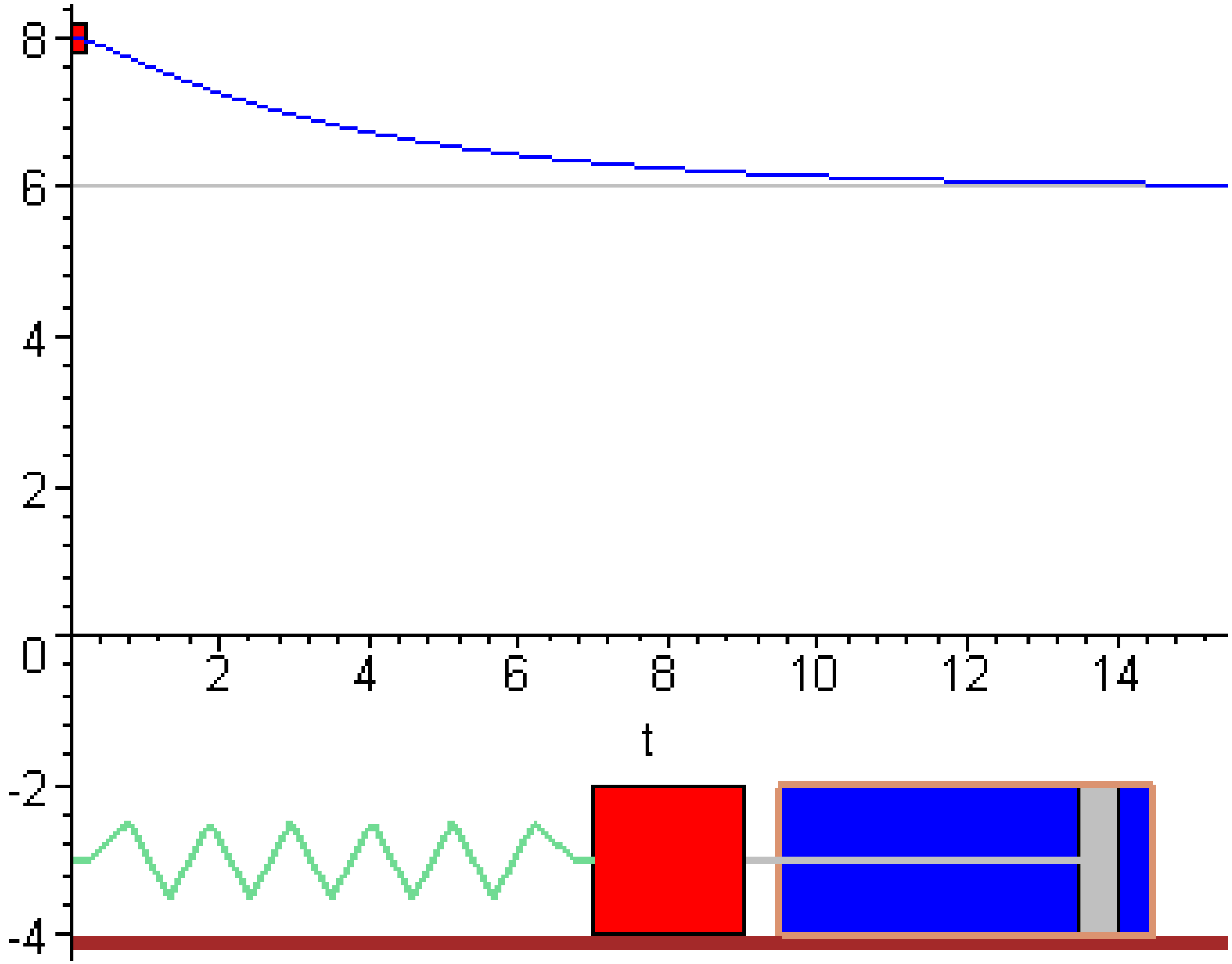
临界阻尼：$x(t)=(c_1+c_2t)\exp(-\dfrac{c}{2m}t)$



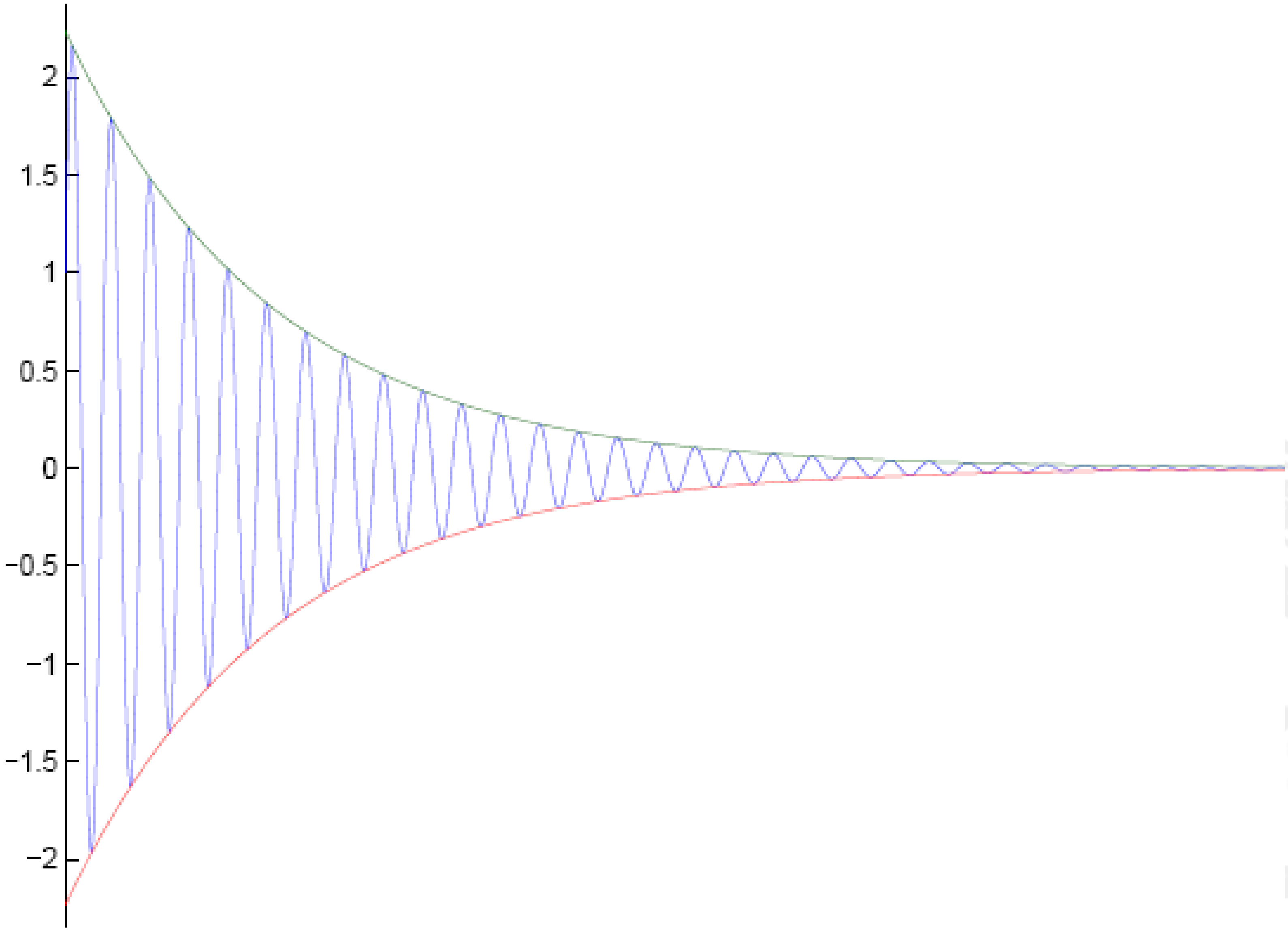
（3）$c^2-4km<0$，两复根，小阻尼

$$x(t)=\exp(-\frac{ct}{2m})[c_1\cos\mu t+c_2\sin\mu t]$$

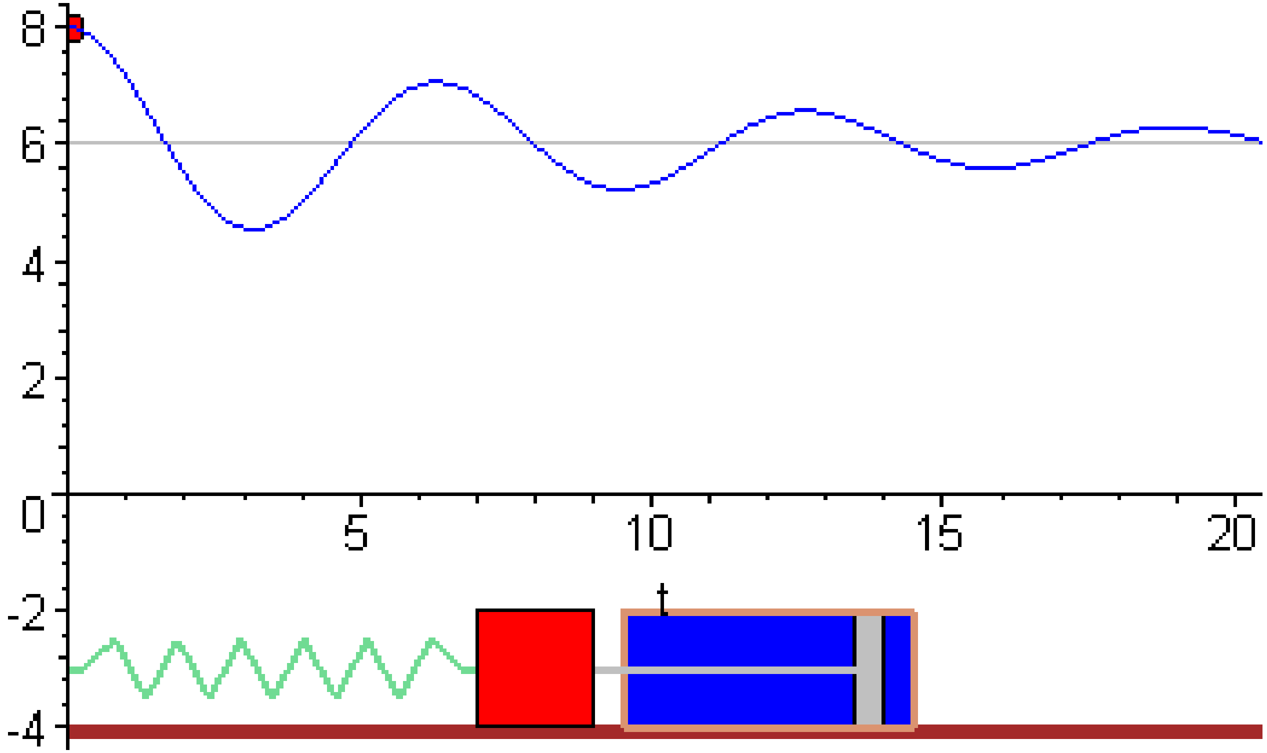
大阻尼情形的动画演示



小阻尼：$x(t)=A\exp(-\frac{ct}{2m})\sin(\mu t+\theta)$



小阻尼情形的动画演示



3 有阻尼的强迫振动——既有空气阻力又有外力

方程 $$m\frac{d^2x}{dt^2}+c\frac{dx}{dt}+kx=F_0\cos\omega t$$

$$\psi(t)=\frac{F_0}{(k-m\omega^2)^2+c^2\omega^2}\left[(k-m\omega^2)\cos\omega t+c\omega\sin\omega t\right]$$

$$=\frac{F_0}{(k-m\omega^2)+c^2\omega^2}\left[(k-m\omega^2)^2+c^2\omega^2\right]^{1/2}\sin(\omega t+\theta)$$

$$=\frac{F_0\sin(\omega t+\theta)}{[(k-m\omega^2)^2+c^2\omega^2]^{1/2}}\qquad \tan\theta=\frac{k-m\omega^2}{c\omega}$$

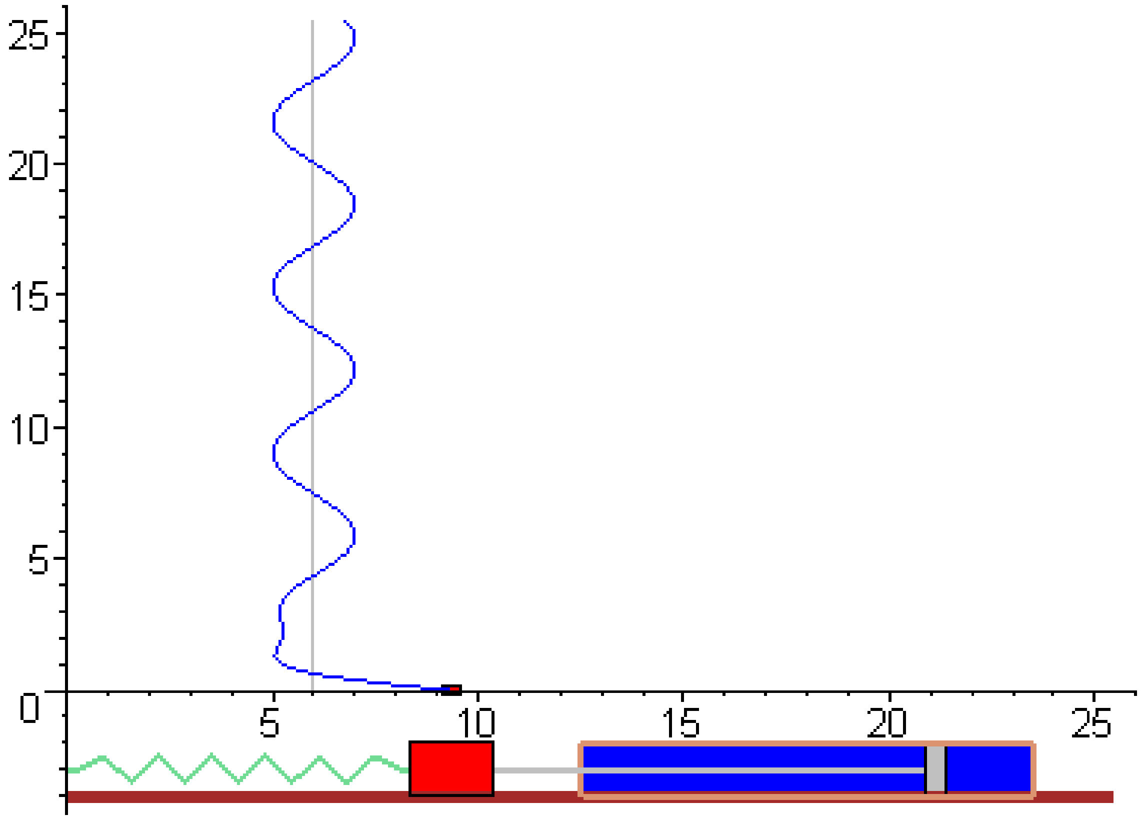
通解为 $$x(t)=\varphi(t)+\frac{F_0\sin(\omega t+\theta)}{[(k-m\omega^2)^2+c^2\omega^2]^{1/2}}$$

当时间比较大时，该解趋于周期运动

这里 $\varphi(t)$ 是方程（3.5.6）对应的齐次方程

周期外力

$$3e^{-t}\cos(2t+\pi/5)+\cos(t+\pi/8)$$



4 无阻尼强迫振动——没有空气阻力而有周期外力

$$\frac{d^2x}{dt^2}+\omega_0^2 x=\frac{F_0}{m}\cos\omega t,\quad \omega_0^2=\frac{k}{m}$$

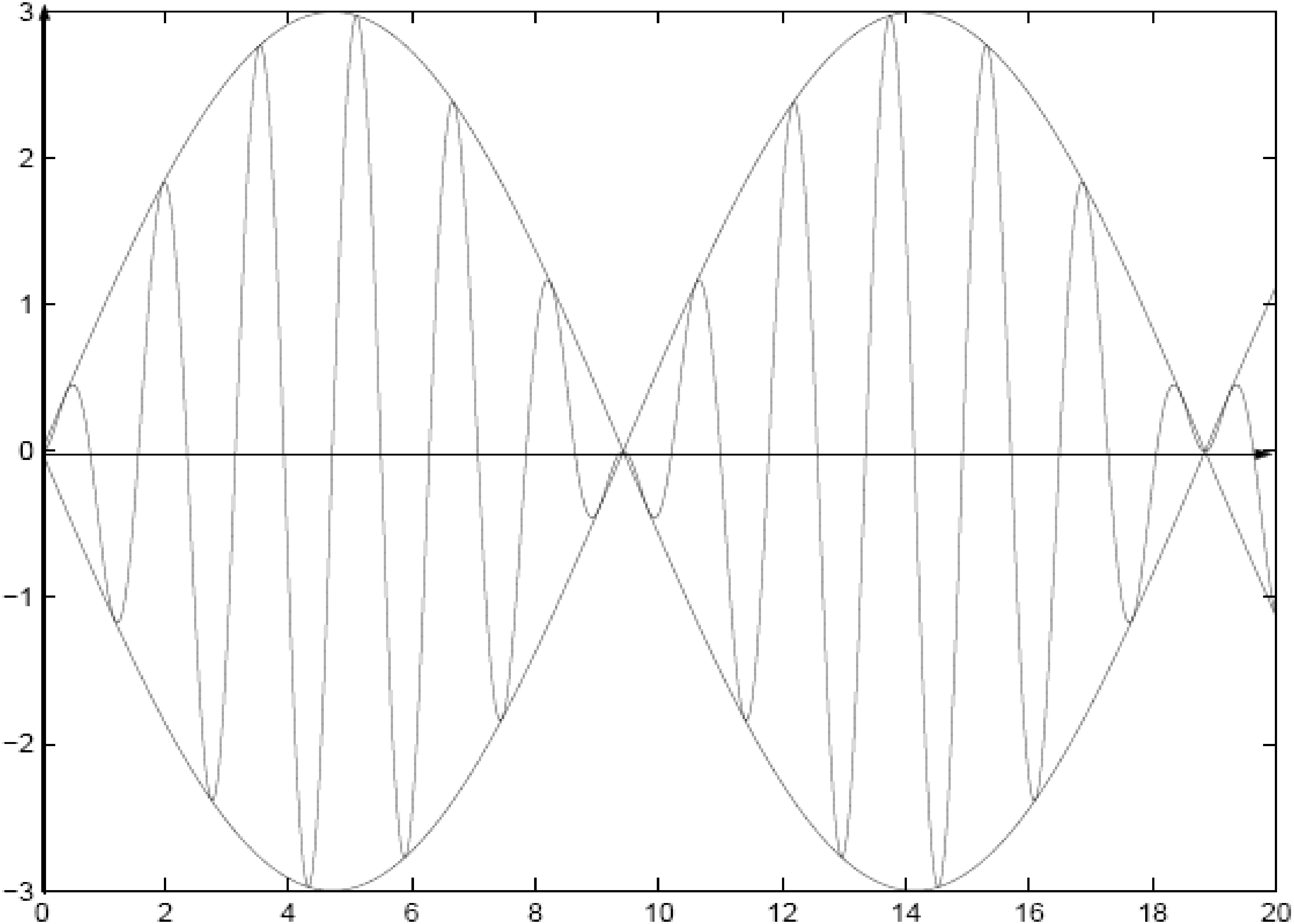
当 $\omega \neq \omega_0$ 时，通解为

$$x(t)=c_1\cos\omega_0 t+c_2\sin\omega_0 t+\frac{F_0}{m(\omega_0^2-\omega^2)}\cos\omega t$$

满足 $x(0)=0,\quad x'(0)=0$ 的解为

$$x(t)=\frac{F_0}{m(\omega_0^2-\omega^2)}(\cos\omega t-\cos\omega_0 t)$$

$$=\frac{2F_0}{m(\omega_0^2-\omega^2)}\sin\frac{(\omega-\omega_0)t}{2}\sin\frac{(\omega+\omega_0)t}{2}$$


3
2
1
0
−1
−2
−3
0
2
4
6
8
10
12
14
16
18
20

当 $\omega=\omega_0$ 时，$\dfrac{d^2x}{dt^2}+\omega_0^2x=\dfrac{F_0}{m}\cos\omega_0 t$

$$x(t)=c_1\cos\omega_0 t+c_2\sin\omega_0 t+\frac{F_0 t}{2m\omega_0}\sin\omega_0 t$$

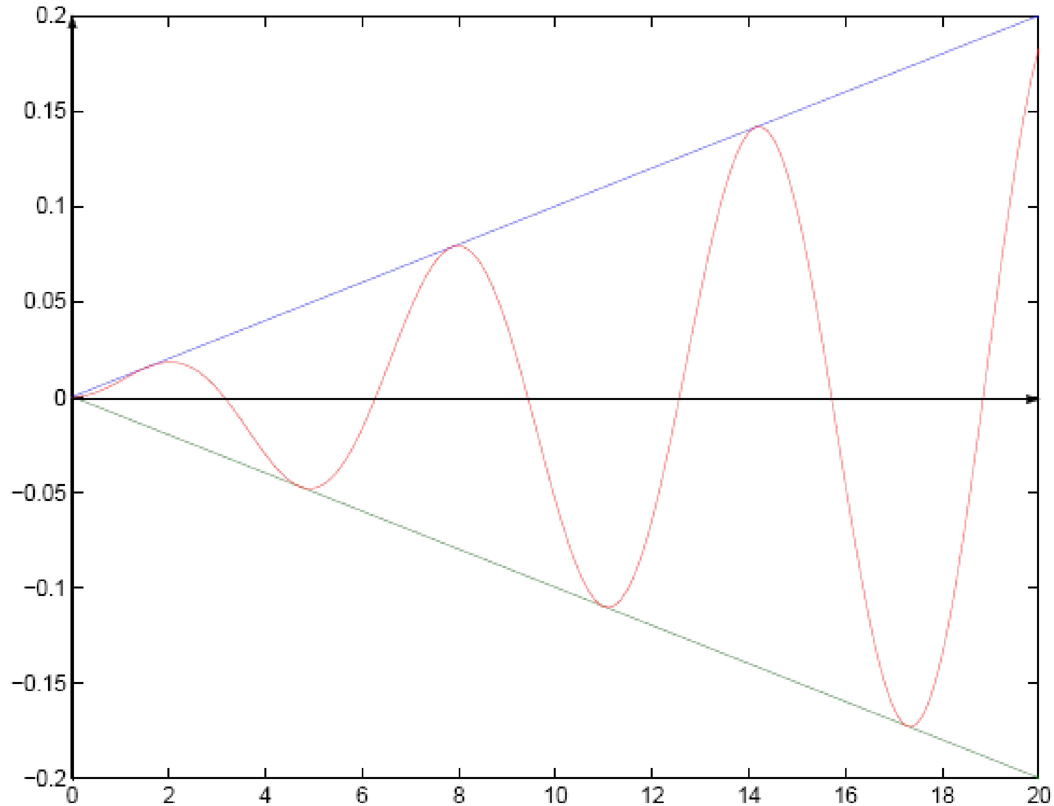
共振现象

有利的方面

音乐

不利的方面

建筑物



阅读材料 The Tacoma Bridge disaster

On July 1, 1940, the Tacoma Narrows Bridge was completed and opened to traffic.

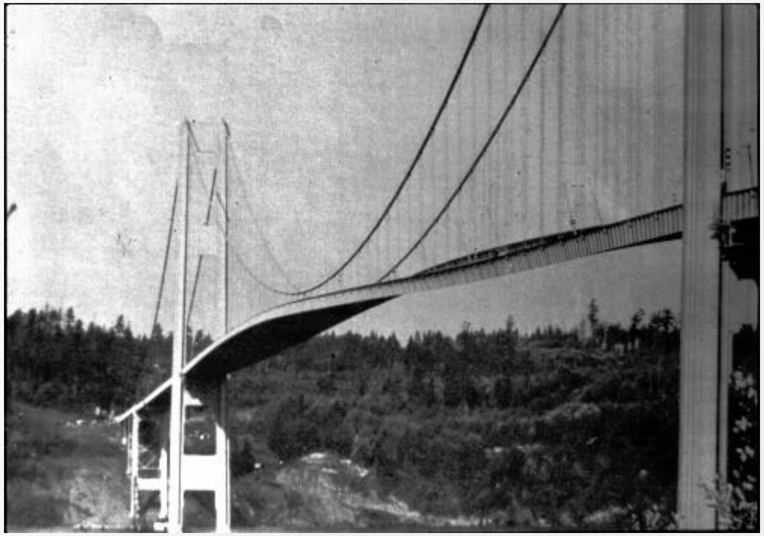
From the day of its opening the bridge began undergoing vertical oscillations, and it soon was nicknamed "Galloping Gertie."

Strange as it may seem, traffic on the bridge increased tremendously as a result of its novel behavior. People came from hundreds of miles in their cars to enjoy the curious thrill of riding over a galloping, rolling bridge.

As each day passed, the authorities became more and more confident of the safety of the bridge.

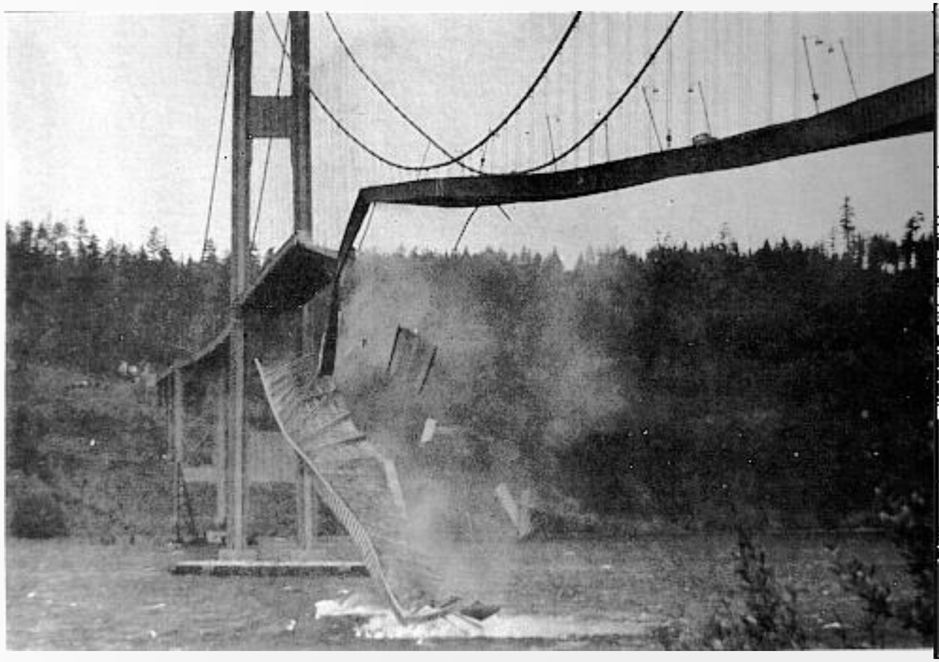
Starting at about 7:00 on the morning of November 7,1940, the bridge began undulating persistently for three hours.

At about 10:00 a.m., the bridge began oscillating wildly. At one moment, one edge of the roadway was twenty-eight feet higher than the other.



At 10:30 a.m., the bridge began cracking, and finally, at 11:10 a.m, the entire bridge came crashing down.

Fortunately, only one car was on the bridge at the time of its failure. A dog went down with the car and the span-the only life lost in the disaster.

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课堂练习

求下面初始值问题的解

$$\frac{d^2x}{dt^2} + x = F(t), \quad x(0) = 0, \quad x'(0) = 0$$

$$F(t) = \begin{cases} 0, & t < 0 \\ 1, & 0 \le t \end{cases}$$

作业　复习题3，8